Making models of polyhedra

A practical pair of lessons: from net to solid, and what a badly folded model teaches that a correct one does not.

LESSON 10–11 | 2 × 40 MIN | GEOMETRY 10, §6 | IGCSE E4.8

LESSON CLOCK

8' | 16' | 30' | 16' | 14' | 6'

What a net must satisfy	8 min
The eleven nets of a cube	16 min
Building — prism, pyramid, tetrahedron	30 min
Surface area from the net	16 min
Checking the models	14 min
Homework	6 min

LEARNING OBJECTIVES

- Draw an accurate net of a prism, a pyramid and a tetrahedron.
- Build the solid from the net and check its face, edge and vertex counts.
- Explain why some arrangements of squares do not fold into a cube.
- Compute the surface area from the net.

TEXTBOOK

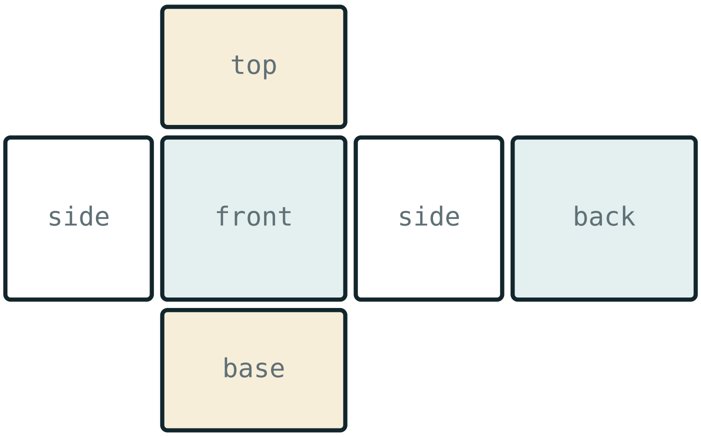
National	pp. 53–60
Cambridge	Core & Extended, pp. 289–293
Workbook	Net templates N1–N5

01 Explanation

What a net has to satisfy

A **net** is the surface of a solid cut along some edges and laid flat. To fold back it must satisfy three conditions:

1. Exactly the right faces, each the right size.
2. Edges that will be joined must be equal in length.
3. The arrangement must not force two faces to overlap.



`surface area = the area of the whole net`

A net and the solid it folds to. Matching edges are the same length.

CONDITION 3 IS THE ONE PEOPLE FORGET

Six squares in a row have the right faces and the right edge lengths and still do not make a cube — the ends meet before the sides close.

The eleven nets of a cube

Of the 35 ways to join six squares edge to edge, exactly **eleven** fold into a cube.
They divide by the length of their longest row:

LONGEST ROW	HOW MANY	DESCRIPTION
4	*6*	a row of four, one square above and one below
3	*4*	two rows, of three and three, offset
2	*1*	a staircase of three pairs

A QUICK TEST

In a cube net, no 2×2 block of squares can appear — such a block would fold two faces onto each other. Any arrangement containing one is not a cube net.

Building, and computing area from the net

Three models, in pairs. Card, ruler, compasses, 5 mm tabs on alternate edges.

MODEL	NET	SURFACE AREA
cube, side a	6 squares	$6a^2$
n-gonal prism	2 bases + a rectangle $P \times h$	$2B + Ph$
regular pyramid	base + n triangles	$B + \frac{1}{2}Pl$
regular tetrahedron, edge a	4 equilateral triangles	$\sqrt{3}\,a^2$

Here B is the base area, P the base perimeter, h the height and l the slant height.

The net makes each formula obvious: the lateral surface of a prism is literally one rectangle.

SLANT HEIGHT IS NOT HEIGHT

For a pyramid the triangles use l, the distance from the apex to the midpoint of a base edge — not the vertical height h. They are related by $l^2 = h^2 + (\frac{\text{base edge}}{2})^2$ for a square base.

02 Worked examples

EXAMPLE 1 **A square-based pyramid has base 10 cm and height 12 cm. Find its surface area.**

1. $l^2 = 12^2 + 5^2 = 169$

   Slant height.

2. $l = 13$

3. Base 100; four triangles $4 \times \frac{1}{2} \times 10 \times 13 = 260$.

4. $100 + 260 = 360$

ANSWER

$360\ cm^2$

EXAMPLE 2 **A triangular prism has an equilateral base of side 6 cm and length 15 cm. Find its surface area.**

1. Base area $\frac{\sqrt{3}}{4} \times 36 = 9\sqrt{3} \approx 15.59$
2. Two bases ≈ 31.18.
3. Lateral: perimeter $18 \times 15 = 270$.
4. Total ≈ 301.2.

ANSWER

≈ $301\ cm^2$

EXAMPLE 3 **Explain why six squares in a single row cannot fold into a cube.**

1. Folding the row wraps four squares round the sides.
2. The fifth returns to the first face’s position.
   Overlap.
3. Two faces are left uncovered.

ANSWER

It violates condition 3 — faces overlap and two are missing

03 Interactive model

Have each pair test one candidate cube net on squared paper before cutting card.

Will it fold?

How many nets of a cube are there?

- *6*
- *8*
- *11*
- *35*

A net containing a 2×2 block of squares:

- folds fine
- never folds to a cube
- makes a cuboid
- makes a pyramid

Surface area of a cube of side 5 cm:

- $25\ cm^2$
- $125\ cm^2$
- $150\ cm^2$
- $100\ cm^2$

The lateral surface of a prism unfolds to:

- a triangle
- one rectangle
- a circle
- a trapezium

A regular tetrahedron’s net is:

- 3 triangles
- 4 triangles
- 4 squares
- 6 triangles

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Net	Yoyma	Развёртка
Fold line	Buklash chizig‘i	Линия сгиба
Tab (flap)	Yopishtirish qismi	Клапан
Surface area	To‘la sirt yuzasi	Площадь полной поверхности
Lateral surface area	Yon sirt yuzasi	Площадь боковой поверхности
Slant height	Apofema	Апофема
Scale of a model	Model masshtabi	Масштаб модели
Template	Andoza	Шаблон
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A net is:

- the volume
- the flattened surface
- a cross-section
- a projection

The slant height of a pyramid is measured to:

- the base centre
- the midpoint of a base edge
- a base vertex
- the apex

Surface area of an n-gonal prism:

- Ph
- $2B + Ph$
- $B + \frac{1}{2}Pl$
- Bh

Six squares in a row fold into:

- a cube
- nothing — they overlap
- a cuboid
- an open box

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Surface area of a cube of side 4 cm

   ANSWER *96 cm²*

2. Surface area of a cube of side 10 cm

   ANSWER *600 cm²*

3. How many faces in a cube net?

   ANSWER *6*

4. How many nets of a cube?

   ANSWER *11*

5. Net of a tetrahedron consists of

   ANSWER *4* equilateral triangles

6. A prism’s lateral net is

   ANSWER one rectangle

7. Slant height of a pyramid with $h = 4$, base edge 6

   ANSWER *5*

Medium · the standard question

7 PROBLEMS

1. Square pyramid, base 8, height 3. Surface area

   ANSWER $144\ cm^2$

2. Cuboid $3 \times 4 \times 5$. Surface area

   ANSWER $94\ cm^2$

3. Triangular prism, right base 3-4-5, length 10. Surface area

   ANSWER $132\ cm^2$

4. Regular tetrahedron edge 6. Surface area

   ANSWER $36\sqrt{3} \approx 62.35\ cm^2$

5. Hexagonal prism, side 2, height 9. Lateral area

   ANSWER 108

6. Cube surface area is 216 cm^2. Find the edge

   ANSWER $6\ cm$

7. Square pyramid, base 12, slant height 10. Surface area

   ANSWER $384\ cm^2$

Hard · stretch and reason

7 PROBLEMS

1. A cube net has a 2×2 block. Prove it cannot fold

   ANSWER Two of the four would land on the same face

2. Square pyramid, base 14, surface area 896. Find the height

   ANSWER 24

3. A cuboid has surface area 208 and edges in the ratio $1{:}2{:}3$

   ANSWER $4 \times 8 \times 12$? check: $S = 352$ — so ratio gives $x \approx 3.1$

4. Regular tetrahedron of surface area $16\sqrt{3}$. Find the edge

   ANSWER 4

5. How many nets has a regular tetrahedron?

   ANSWER 2

6. A prism has surface area $2B + Ph$. What happens as $h \to 0$?

   ANSWER It flattens to twice the base

7. Design a net for an oblique prism and say why it is harder

   ANSWER The lateral faces are parallelograms of differing shapes

07 Homework

Homework — 4 tasks

Bring the three finished models to the next lesson — they are used for sections.

1. Draw and cut two different nets of a cube, and fold both.
2. Draw the net of a square pyramid with base 8 cm and height 3 cm, and find its surface area.
3. Draw the net of a regular tetrahedron of edge 6 cm and find its surface area.
4. Find an arrangement of six squares that is not a cube net and explain in one sentence why it fails.